

Beyond User Experience: Technical and Contextual Metrics for Large Language Models in Extended Reality

Yiliu Tang
yiliut2@illinois.edu
University of Illinois
Urbana-Champaign
School of Information Sciences
Champaign, Illinois, USA

Jason Situ
junsitu2@illinois.edu
University of Illinois
Urbana-Champaign
Siebel School of Computing and Data
Science
Champaign, Illinois, USA

Yun Huang
yunhuang@illinois.edu
University of Illinois
Urbana-Champaign
School of Information Sciences
Champaign, Illinois, USA

Abstract

Spatial Computing involves interacting with the physical world through spatial data manipulation, closely linked with Extended Reality (XR), which includes Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR). Large Language Models (LLMs) significantly enhance XR applications by improving user interactions through natural language understanding and content generation. Typical evaluations of these applications focus on user experience (UX) metrics, such as task performance, user satisfaction, and psychological assessments, but often neglect the technical performance of the LLMs themselves. This paper identifies significant gaps in current evaluation practices for LLMs within XR environments, attributing them to the novelty of the field, the complexity of spatial contexts, and the multimodal nature of interactions in XR. To address these gaps, the paper proposes specific metrics tailored to evaluate LLM performance in XR contexts, including spatial contextual awareness, coherence, proactivity, multimodal integration, hallucination, and question-answering accuracy. These proposed metrics aim to complement existing UX evaluations, providing a comprehensive assessment framework that captures both the technical and user-centric aspects of LLM performance in XR applications. The conclusion underscores the necessity for a dual-focused approach that combines technical and UX metrics to ensure effective and user-friendly LLM-integrated XR systems.

CCS Concepts

• **Human-centered computing** → **HCI theory, concepts and models.**

Keywords

Spatial Computing; Extended Reality; Virtual Reality; Augmented Reality; Mixed Reality; Large Language Models; User Experience; Evaluation Metrics

ACM Reference Format:

Yiliu Tang, Jason Situ, and Yun Huang. 2024. Beyond User Experience: Technical and Contextual Metrics for Large Language Models in Extended Reality. In *Companion of the 2024 ACM International Joint Conference on*

Pervasive and Ubiquitous Computing (UbiComp Companion '24), October 5–9, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 4 pages. <https://doi.org/10.1145/3675094.3678995>

1 Introduction

Spatial Computing refers to the technologies and techniques that interact with the physical world through the manipulation of spatial data [8]. It focuses on the interaction and integration between digital and physical spaces, emphasizing spatial relationships and context [4, 22]. Extended Reality (XR) is a key component of spatial computing. XR is an umbrella term encompassing technologies that merge the virtual and real worlds, including Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR) [11]. VR refers to a fully immersive digital environment where users interact with virtual spaces that are simulated, cutting them off from the real world [1, 38]. AR refers to overlaying digital content such as images or information onto the real world through intelligent devices such as smartphones, AR glasses, and AR projectors [3]. Lastly, MR experiences take AR a step further, allowing virtual objects to coexist and interact with the real world, creating a seamless blend between the virtual and physical [28].

Large Language Models (LLMs) are machine learning models trained on massive amounts of text data [37]. They have found significant applications in spatial computing, particularly within XR environments. Their ability to understand and generate human-like text makes them valuable for enhancing user interactions in immersive experiences [6, 14, 15, 32]. Some of the application domains of LLMs incorporated into XR include education, healthcare, entertainment and tourism [2, 16, 30, 33].

The evaluation of XR-based applications has predominantly focused on user experience (UX) metrics from both behavioral and perceptual perspectives. These metrics often measure aspects like user satisfaction, perceived usability of the system, cognitive load, and emotional assessment [16, 19, 26, 33]. For instance, in a language learning application in VR settings, evaluations might focus on how users feel about their language skills improvement or their overall satisfaction with the learning experience [17]. However, these evaluations frequently overlook the technical performance of the LLM itself. These include the accuracy, relevance, and safety of the responses generated by the LLM during the interview simulations.



This work is licensed under a Creative Commons Attribution International 4.0 License.

UbiComp Companion '24, October 5–9, 2024, Melbourne, VIC, Australia
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-1058-2/24/10
<https://doi.org/10.1145/3675094.3678995>

2 Current Evaluation Metrics in XR Environments with LLM Integration

We conducted a literature survey on the use of LLMs in XR environments. After reviewing 89 papers, we identified the following categories of evaluation metrics.

2.1 Task Performance

This category assesses how effectively users can perform tasks in XR settings. While task performance is often associated with how effectively an LLM contributes to the completion of a task, it does not directly measure the LLM's performance.

- Time of Task Completion [34]: This metric measures the time users take to complete designated tasks within the XR environment. It is a direct indicator of task efficiency. Monitoring task completion time helps developers understand how different design choices impact user efficiency and effectiveness.
- Accuracy of Physical Interactions [34]: This metric evaluates how accurately users can interact with virtual objects or navigate through virtual spaces. It is important in training simulations or educational applications where precise interactions are necessary for skill development or knowledge acquisition. Accuracy not only reflects the user's ability to perform tasks but also indicates how well the XR environment replicates real-world conditions and supports user goals, providing a key measure of system fidelity and user interface design effectiveness.

2.2 User Satisfaction and Interface Usability

This category evaluates how users perceive and interact with XR systems, focusing on their overall satisfaction and the usability of the interface.

- System Usability Scale (SUS) [25]: It consists of a 10-item questionnaire with five response options, from "Strongly agree" to "Strongly disagree," providing subjective assessments of usability. It enables developers to identify specific aspects of the user interface that may require improvements to enhance ease of use and user satisfaction.
- NASA Task Load Index (TLX) [34]: The NASA TLX measures perceived workload to assess the cognitive load involved in completing tasks. It evaluates multiple dimensions such as mental demand, physical demand, temporal demand, performance, effort, and frustration.
- User-Rated Need Satisfaction and Engagement [17]: This involves direct feedback from users regarding the extent to which the system meets their needs and expectations. Engagement levels are often assessed using Likert scales to quantify how engaging and interactive the user finds the XR application.

2.3 Psychological and Emotional Assessment in XR Environments with LLM Integration

This category examines the methods used to assess the emotional and psychological state of users engaging with LLM-driven XR

interfaces. These metrics are crucial for understanding the emotional depth and psychological comfort users experience during their interactions.

- State-Trait Anxiety Inventory (STAI) [20]: It provides a clear distinction between state anxiety (a temporary condition in response to a situation) and trait anxiety (more long-term, chronic anxiety). It's particularly useful in settings where the emotional response can significantly impact the user's experience or performance, such as in high-stress training simulations or therapeutic scenarios.
- Self-assessment Manikin (SAM) [20]: SAM is a non-verbal pictorial assessment technique that measures pleasure, arousal, and dominance felt by individuals in response to a variety of stimuli.

2.4 Conversational Analysis

Metrics Focus: This category emphasizes the use of conversational metrics to obtain insights into how users perceive and interact within XR environments.

- Thematic Analysis of Transcribed Interviews [29]: Thematic analysis involves identifying, analyzing, and reporting patterns (themes) within data to uncover user attitudes, feelings, and detailed feedback on specific aspects of the XR environment, such as usability, engagement, and the emotional impact of interactions.
- Conversational Analysis [39]: These metrics include measurements of conversation content, recall, length, breadth, and sentiment. They are used to quantitatively evaluate the quality of conversations within XR systems.

3 Gaps in Current Metrics for LLMs in XR

Integrating LLMs into XR applications is still at its early stage. We noticed a lack of LLM evaluation metrics in LLM-enabled XR applications. Most existing evaluation frameworks are designed for either LLMs or spatial computing independently, not accounting for the unique challenges and requirements when these technologies are combined. There may be two additional reasons.

3.1 Complexity of Spatial Contexts

Evaluating LLMs in spatial environments requires considering a range of factors that traditional NLP metrics do not fully capture. This includes spatial understanding, context awareness, and the ability to interact with 3D environments [10, 27]. For example, traditional metrics like BLEU [23] or ROUGE [18] score might measure the quality of text generation in terms of linguistic accuracy but fail to account for how well the generated text logically aligns with a spatial context or aids in navigating a virtual environment.

3.2 Multimodal Interactions

LLMs in spatial computing often interact with other modalities, such as visual and haptic feedback, which adds layers of complexity to the evaluation process [24]. The performance of LLMs in these environments cannot be accurately assessed using metrics that only consider text output. New metrics need to be developed to evaluate the effectiveness of LLMs in integrating and interacting with visual

elements, spatial cues, and user inputs within XR environments [31].

4 Proposed Metrics for LLM Evaluations in XR

To address the gaps in current evaluation practices, we propose a set of metrics tailored for assessing LLM performance in XR environments, complementing existing UX evaluations. Some of these proposed metrics find their root in traditional LLM evaluation procedures, but carry extended meaning in the XR context.

- **Spatial Contextual Awareness** [7, 9]: This metric evaluates the accuracy of the LLM's responses in relation to spatial context and tasks within the XR environment. This can include tasks such as generating directions to guide a user through a virtual environment, providing context-aware information about objects, and responding appropriately to spatial cues.
- **Coherence** [21]: This metric measures the LLM's ability to maintain logical and contextually consistent conversations over extended interactions within an immersive environment.
- **Proactivity** [5]: This metric measures the LLM's ability to proactively provide useful information or suggestions in XR without explicit user prompts.
- **Multimodal Integration** [35, 36]: Evaluates how well the LLM integrates and responds to multimodal inputs, such as voice, gestures, and visual cues.
- **Hallucination** [12]: Hallucination evaluates the propensity of the LLM to generate factually incorrect or illogical statements.
- **Question-Answering Accuracy** [13]: This metric assesses how effectively the LLM handles direct user inquiries within XR environments.

5 Discussion

In summary, while current evaluation practices for LLMs in XR applications emphasize user experience, there is a lack of comprehensive metrics that assess the accuracy and relevance of LLM-generated responses in XR. While the UX metrics capture users' perceived experience, new metrics need to be developed to evaluate the LLM's capabilities. Combining these metrics will provide a holistic view of the LLM's performance in the context of XR.

References

- [1] Christoph Anthes, Rubén Jesús García-Hernández, Markus Wiedemann, and Dieter Kranzlmüller. 2016. State of the art of virtual reality technology. In *2016 IEEE aerospace conference*. IEEE, 1–19.
- [2] Dennis Ayre, Carolyn Dougherty, and Yitong Zhao. 2023. Implementation of an Artificial Intelligence (AI) Instructional Support System in a Virtual Reality (VR) Thermal-Fluids Laboratory. In *ASME International Mechanical Engineering Congress and Exposition*, Vol. 87653. American Society of Mechanical Engineers, V008T09A029.
- [3] Ronald T Azuma. 1997. A survey of augmented reality. *Presence: teleoperators & virtual environments* 6, 4 (1997), 355–385.
- [4] S Balakrishnan, M Syed Shahul Hameed, K Venkatesan, and G Aswin. 2021. Interaction of spatial computing in augmented reality. In *2021 7th International Conference on Advanced Computing and Communication Systems (ICACCS)*, Vol. 1. IEEE, 1900–1904.
- [5] Daniel Billsus, David M Hilbert, and Dan Maynes-Aminzade. 2005. Improving proactive information systems. In *Proceedings of the 10th international Conference on intelligent User interfaces*. 159–166.
- [6] Yupeng Chang, Xu Wang, Jindong Wang, Yuan Wu, Linyi Yang, Kaijie Zhu, Hao Chen, Xiaoyuan Yi, Cunxiang Wang, Yidong Wang, et al. 2024. A survey on evaluation of large language models. *ACM Transactions on Intelligent Systems and Technology* 15, 3 (2024), 1–45.
- [7] Guanling Chen and David Kotz. 2000. A survey of context-aware mobile computing research. (2000).
- [8] Simon Greenwold. 2003. Spatial computing. *Massachusetts Institute of Technology, Master* (2003).
- [9] Andy Harter, Andy Hopper, Pete Steggles, Andy Ward, and Paul Webster. 1999. The anatomy of a context-aware application. In *Proceedings of the 5th annual ACM/IEEE international conference on Mobile computing and networking*. 59–68.
- [10] Aziz Hasanov, Teemu H Laine, and Tae-Sun Chung. 2019. A survey of adaptive context-aware learning environments. *Journal of Ambient Intelligence and Smart Environments* 11, 5 (2019), 403–428.
- [11] Teresa Hirzle, Florian Müller, Fiona Draxler, Martin Schmitz, Pascal Knierim, and Kasper Hornbæk. 2023. When xr and ai meet—a scoping review on extended reality and artificial intelligence. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–45.
- [12] Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. 2023. Survey of hallucination in natural language generation. *Comput. Surveys* 55, 12 (2023), 1–38.
- [13] Zhengbao Jiang, Frank F Xu, Jun Araki, and Graham Neubig. 2020. How can we know what language models know? *Transactions of the Association for Computational Linguistics* 8 (2020), 423–438.
- [14] Enkelejd Kasneci, Kathrin Seßler, Stefan Küchemann, Maria Bannert, Daryna Dementieva, Frank Fischer, Urs Gasser, Georg Groh, Stephan Günemann, Eyke Hüllermeier, et al. 2023. ChatGPT for good? On opportunities and challenges of large language models for education. *Learning and individual differences* 103 (2023), 102274.
- [15] John Kirchenbauer, Jonas Geiping, Yuxin Wen, Jonathan Katz, Ian Miers, and Tom Goldstein. 2023. A watermark for large language models. In *International Conference on Machine Learning*. PMLR, 17061–17084.
- [16] Hyungmin Lee, Chen-Chun Hsia, Aleksandr Tsoy, Sungmin Choi, Hanchao Hou, and Shiguang Ni. 2023. VisionARY: Exploratory research on Contextual Language Learning using AR glasses with ChatGPT. In *Proceedings of the 15th Biannual Conference of the Italian SIGCHI Chapter*. 1–6.
- [17] Hyungmin Lee, Chen-Chun Hsia, Aleksandr Tsoy, Sungmin Choi, Hanchao Hou, and Shiguang Ni. 2023. VisionARY: Exploratory research on Contextual Language Learning using AR glasses with ChatGPT. In *Proceedings of the 15th Biannual Conference of the Italian SIGCHI Chapter (CHIItaly '23)*. Association for Computing Machinery, New York, NY, USA, 1–6. <https://doi.org/10.1145/3605390.3605400>
- [18] Chin-Yew Lin. 2004. Rouge: A package for automatic evaluation of summaries. In *Text summarization branches out*. 74–81.
- [19] Jose Llanes-Jurado, Lucía Gómez-Zaragoza, Maria Eleonora Minissi, Mariano Alcañiz, and Javier Marin-Morales. 2024. Developing conversational Virtual Humans for social emotion elicitation based on large language models. *Expert Systems with Applications* 246 (2024), 123261.
- [20] Jose Llanes-Jurado, Lucía Gómez-Zaragoza, Maria Eleonora Minissi, Mariano Alcañiz, and Javier Marin-Morales. 2024. Developing conversational Virtual Humans for social emotion elicitation based on large language models. *Expert Systems with Applications* 246 (July 2024), 123261. <https://doi.org/10.1016/j.eswa.2024.123261>
- [21] David Newman, Jey Han Lau, Karl Grieser, and Timothy Baldwin. 2010. Automatic evaluation of topic coherence. In *Human language technologies: The 2010 annual conference of the North American chapter of the association for computational linguistics*. 100–108.
- [22] Erin Pangilinan, Steve Lukas, and Vasanth Mohan. 2019. *Creating augmented and virtual realities: theory and practice for next-generation spatial computing*. "O'Reilly Media, Inc".
- [23] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. 2002. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*. 311–318.
- [24] Ismo Rakkolainen, Ahmed Farooq, Jari Kangas, Jaakko Hakulinen, Jussi Rantala, Markku Turunen, and Roope Raisamo. 2021. Technologies for multimodal interaction in extended reality—a scoping review. *Multimodal Technologies and Interaction* 5, 12 (2021), 81.
- [25] Devanshu Saindane, Sunny Prakash Prajapati, and Syaamantak Das. [n. d.]. Converting Physical Textbooks into Interactive and Immersive 'Phygital' Textbooks: A Proposed System Architecture Design for Textbook Companion Apps. ([n. d.]).
- [26] Devanshu Saindane, Sunny Prakash Prajapati, and Syaamantak Das. 2023. Converting Physical Textbooks into Interactive and Immersive 'Phygital' Textbooks: A Proposed System Architecture Design for Textbook Companion Apps.. In *iTextbooks@ AIED*. 88–93.
- [27] Marc A Schnabel and Thomas Kvan. 2003. Spatial understanding in immersive virtual environments. *International Journal of Architectural Computing* 1, 4 (2003), 435–448.
- [28] Maximilian Speicher, Brian D Hall, and Michael Nebeling. 2019. What is mixed reality?. In *Proceedings of the 2019 CHI conference on human factors in computing systems*. 1–15.

- [29] B.M.R. Spiegel, O. Liran, A. Clark, J.S. Samaan, C. Khalil, R. Chernoff, K. Reddy, and M. Mehra. 2024. Feasibility of combining spatial computing and AI for mental health support in anxiety and depression. *npj Digital Medicine* 7, 1 (2024). <https://doi.org/10.1038/s41746-024-01011-0>
- [30] Brennan MR Spiegel, Omer Liran, Allistair Clark, Jamil S Samaan, Carine Khalil, Robert Chernoff, Kavya Reddy, and Muskaan Mehra. 2024. Feasibility of combining spatial computing and AI for mental health support in anxiety and depression. *NPJ Digital Medicine* 7, 1 (2024), 22.
- [31] Kay M Stanney, Hannah Nye, Sam Haddad, Kelly S Hale, Christina K Padron, and Joseph V Cohn. 2021. Extended reality (XR) environments. *Handbook of human factors and ergonomics* (2021), 782–815.
- [32] Arun James Thirunavukarasu, Darren Shu Jeng Ting, Kabilan Elangovan, Laura Gutierrez, Ting Fang Tan, and Daniel Shu Wei Ting. 2023. Large language models in medicine. *Nature medicine* 29, 8 (2023), 1930–1940.
- [33] Fang Xu, Tri Nguyen, and Jing Du. 2024. Augmented reality for maintenance tasks with ChatGPT for automated text-to-action. *Journal of Construction Engineering and Management* 150, 4 (2024), 04024015.
- [34] Fang Xu, Tri Nguyen, and Jing Du. 2024. Augmented Reality for Maintenance Tasks with ChatGPT for Automated Text-to-Action. *Journal of Construction Engineering and Management* 150, 4 (April 2024), 04024015. <https://doi.org/10.1061/JCEMD4.COENG-14142> Publisher: American Society of Civil Engineers.
- [35] Shukang Yin, Chaoyou Fu, Sirui Zhao, Ke Li, Xing Sun, Tong Xu, and Enhong Chen. 2023. A survey on multimodal large language models. *arXiv preprint arXiv:2306.13549* (2023).
- [36] Weihao Yu, Zhengyuan Yang, Linjie Li, Jianfeng Wang, Kevin Lin, Zicheng Liu, Xinchao Wang, and Lijuan Wang. 2023. Mm-vet: Evaluating large multimodal models for integrated capabilities. *arXiv preprint arXiv:2308.02490* (2023).
- [37] Wayne Xin Zhao, Kun Zhou, Junyi Li, Tianyi Tang, Xiaolei Wang, Yupeng Hou, Yingqian Min, Beichen Zhang, Junjie Zhang, Zican Dong, et al. 2023. A survey of large language models. *arXiv preprint arXiv:2303.18223* (2023).
- [38] Qingxiao Zheng, Shengyang Xu, Lingqing Wang, Yiliu Tang, Rohan C Salvi, Guo Freeman, and Yun Huang. 2023. Understanding safety risks and safety design in social VR environments. *Proceedings of the ACM on Human-Computer Interaction* 7, CSCW1 (2023), 1–37.
- [39] Jiarui Zhu, Radha Kumaran, Chengyuan Xu, and Tobias Höllerer. 2023. Free-form Conversation with Human and Symbolic Avatars in Mixed Reality. In *2023 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*. IEEE, 751–760.